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1. Define a string "PYTHON IS FUN" and print every 3rd character of the string.

```
# Define the string
text = "PYTHON IS FUN"

# Print every 3rd character
print(text[::3])
```

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2. Create a list containing 5 integers. In the list, modify each even index element by adding it with the previous index element. Print the list before and after modification.

```
# Create a list of 5 integers
numbers = [10, 20, 30, 40, 50]

# Print original list
print("Before modification:", numbers)

# Modify even index elements (except index 0)
for i in range(1, len(numbers)):
    if i % 2 == 0:
        numbers[i] = numbers[i] + numbers[i - 1]

# Print modified list
print("After modification:", numbers)
```

Before modification: [10, 20, 30, 40, 50]
After modification: [10, 20, 50, 40, 90]

3. Create a tuple with 3 random integers. Now try to swap the 1st and the last elements. Print the result.

```
import random

# Create a tuple with 3 random integers
t = (random.randint(1, 100), random.randint(1, 100), random.randint(1, 100))

print("Original tuple:", t)

# Tuples are immutable, so convert to list first
temp = list(t)

# Swap first and last elements
temp[0], temp[-1] = temp[-1], temp[0]

# Convert back to tuple
t_swapped = tuple(temp)

print("After swapping:", t_swapped)
```

Original tuple: (45, 91, 54)
After swapping: (54, 91, 45)

4. Create a dictionary with key/values: 0/'We', '1'/'Love', 'Last'/'PYTHON PROGRAMMING'. Try to print each value specifying the keys.

```
# Create the dictionary
my_dict = {0: 'We', '1': 'Love', 'Last': 'PYTHON PROGRAMMING'}

# Print values using their keys
print(my_dict[0])
print(my_dict['1'])
print(my_dict['Last'])
```

We
Love
PYTHON PROGRAMMING

5. Create a list of integers from 1 to 50. Print all the values that are divisible by 3 and not divisible by 2.

```
# Create list from 1 to 50
numbers = list(range(1, 51))

# Print numbers divisible by 3 and not divisible by 2
for num in numbers:
    if num % 3 == 0 and num % 2 != 0:
        print(num, end=" ")
```

3 9 15 21 27 33 39 45

6. Create a list X of odd numbers between 10 and 30 and print them. Create a new list Y which contains cubic of each element of X.

```
# Create list X (odd numbers between 10 and 30)
X = [num for num in range(10, 31) if num % 2 != 0]
```

```
print("List X:", X)
```

```
# Create list Y (cube of each element in X)
Y = [num**3 for num in X]
```

```
print("List Y (cubes):", Y)
```

List X: [11, 13, 15, 17, 19, 21, 23, 25, 27, 29]

List Y (cubes): [1331, 2197, 3375, 4913, 6859, 9261, 12167, 15625, 19683, 24389]